"Deadlock"

if two or more pr

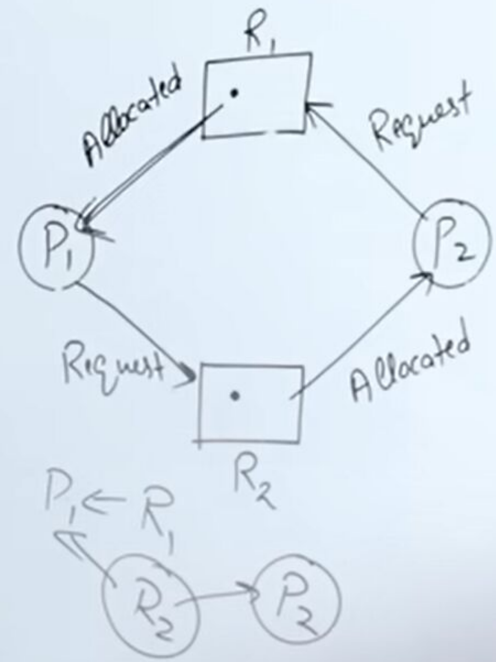
on happening of some e.

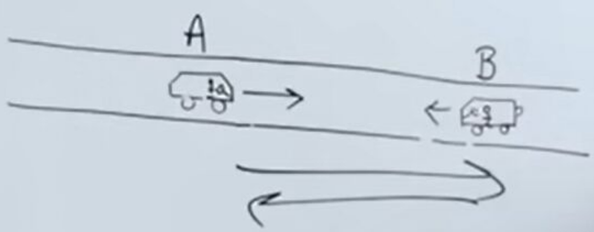
never happens, then we say

are involved in deadlock then that state

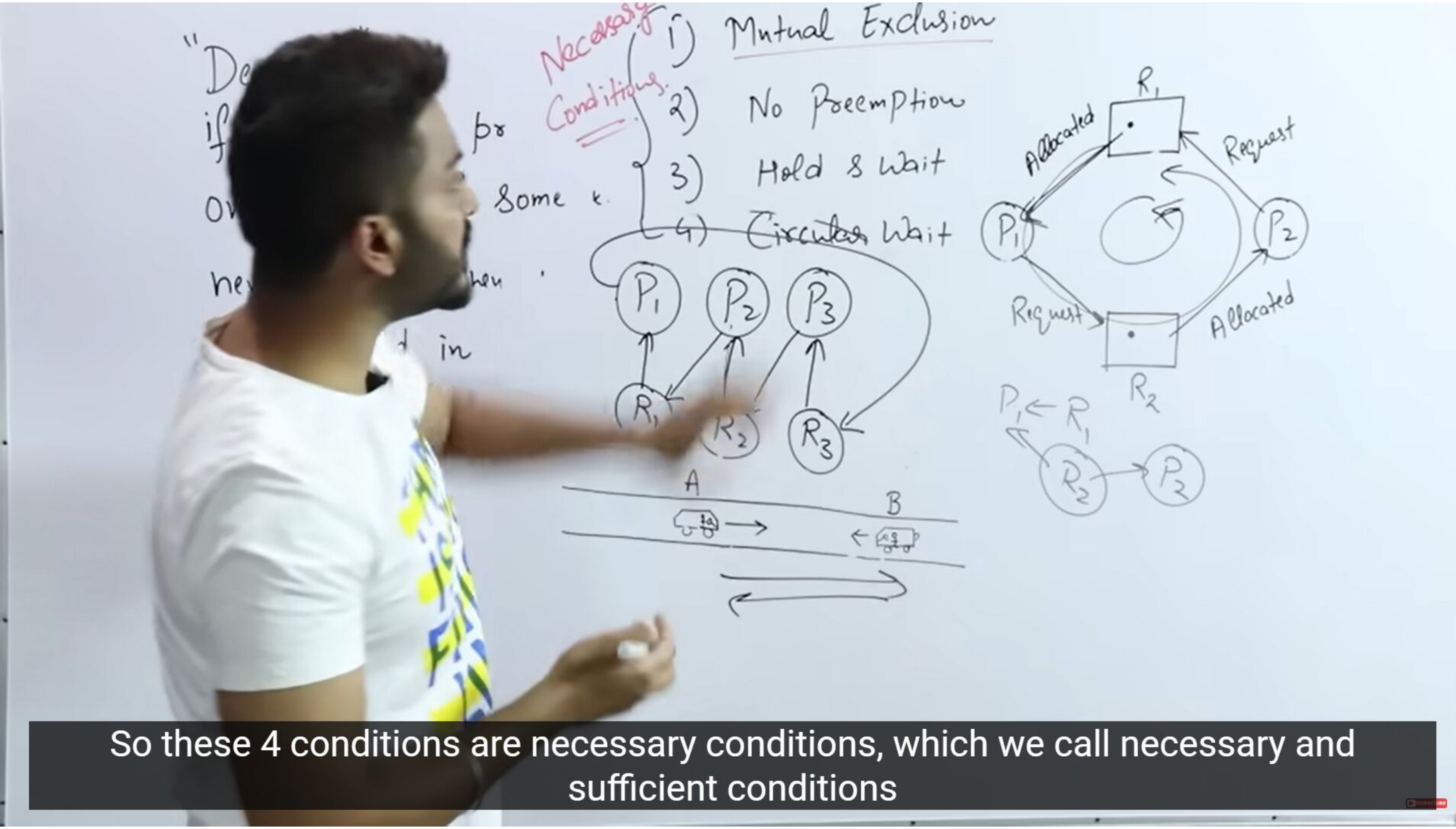
is called Deadlock.

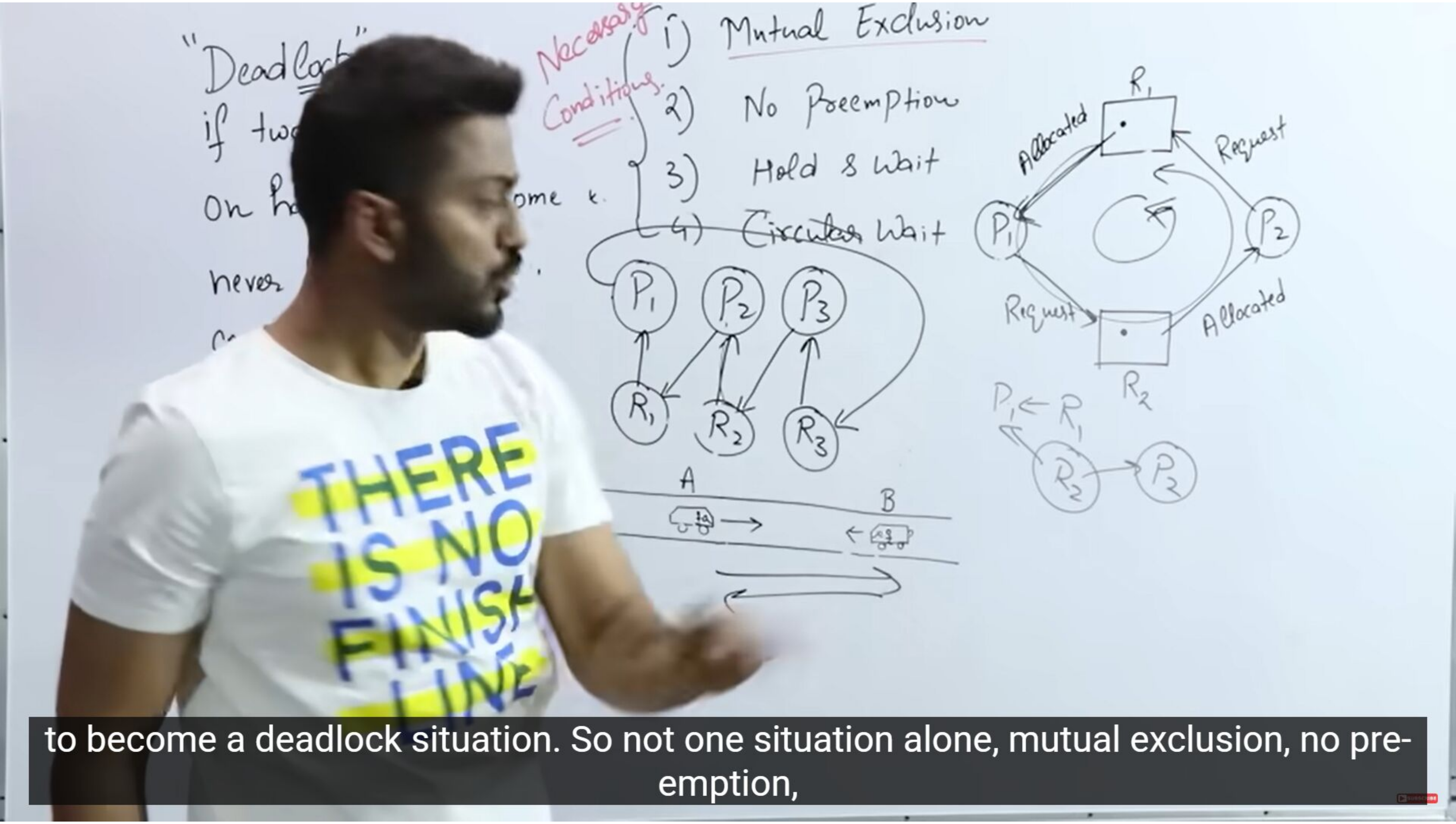
1) Mutual Exclusion
2) No Preemption
3) Hold & Wait
4) Circular Wait





These 4 are necessary conditions for the deadlock.


Necessary Conditions
1) Mutual Exclusion
2) No Preemption
3) Hold & Wait
4) Circular Wait
P1 P2 P3
R1 R2 R3
A
B
R1
R2
Allocated
Request
Request
Allocated
P1
P2
P1 ← R1
R2
P2
So these 4 conditions are necessary conditions, which we call necessary and sufficient conditions



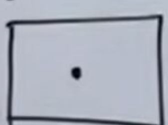
Necessary Conditions
1) Mutual Exclusion
2) No Preemption
3) Hold & Wait
4) Circular Wait
P1
P2
P3
R1
R2
R3
A
B
R1
R2
Allocated
Request
Request
Allocated
P1
P2
P1 ← R1
R2
P2
THERE IS NO FINISH LINE
to become a deadlock situation. So not one situation alone, mutual exclusion, no pre-emption,

...urce Allocation graph (RAG)

...ertex

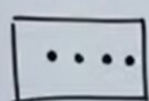
Resource Vertex

Single Instance



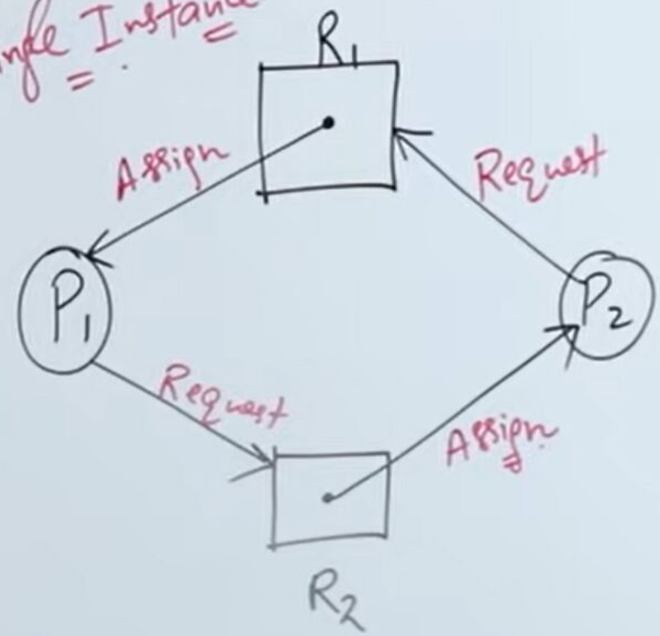
e.g CPU, Monitor

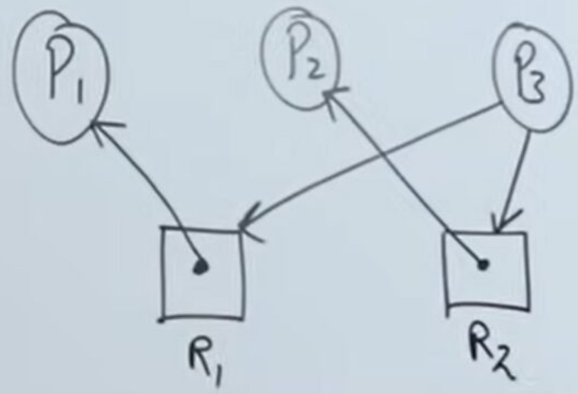
Multi Ins.

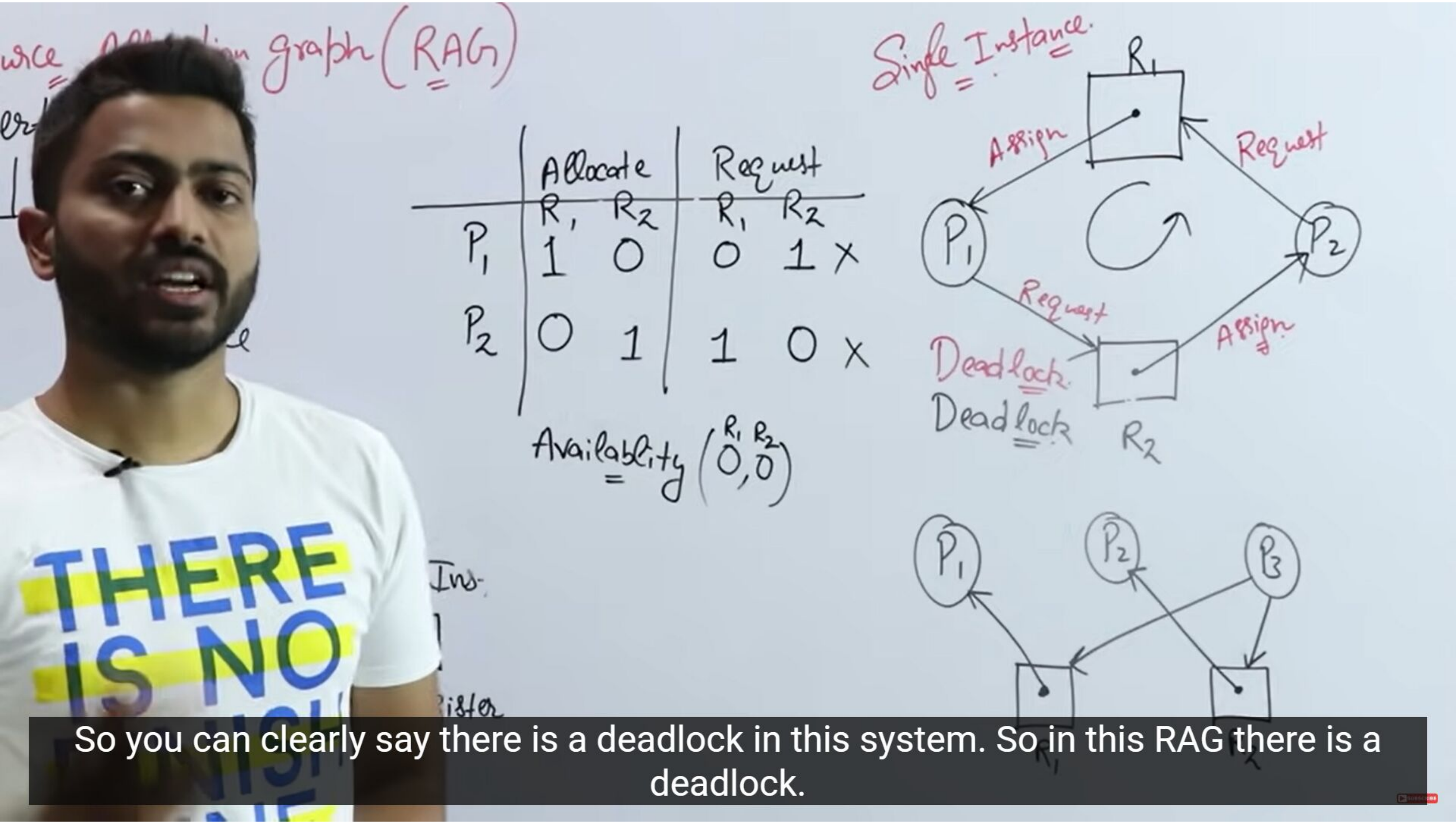


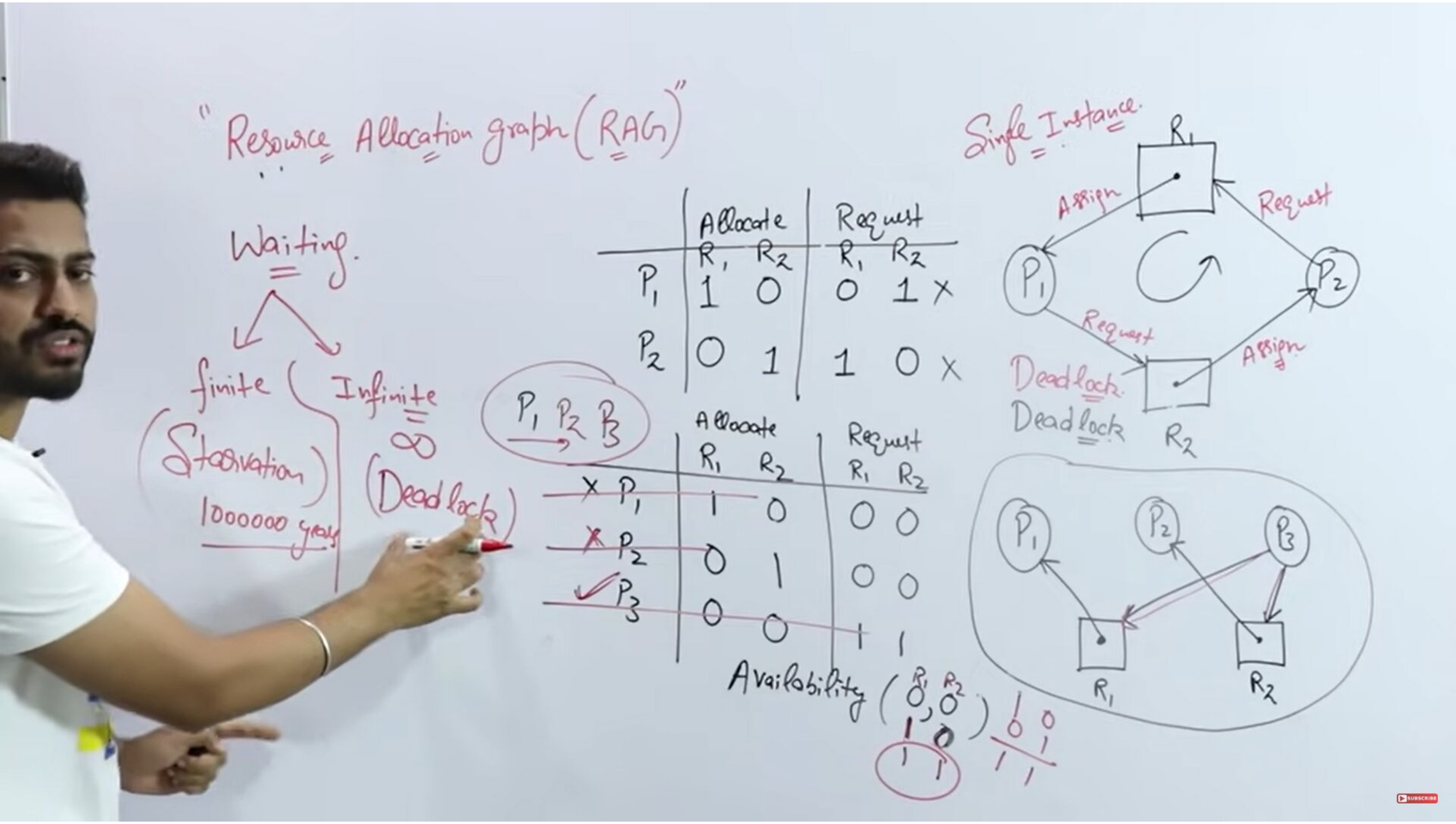
e.g. Register

Single Instance.

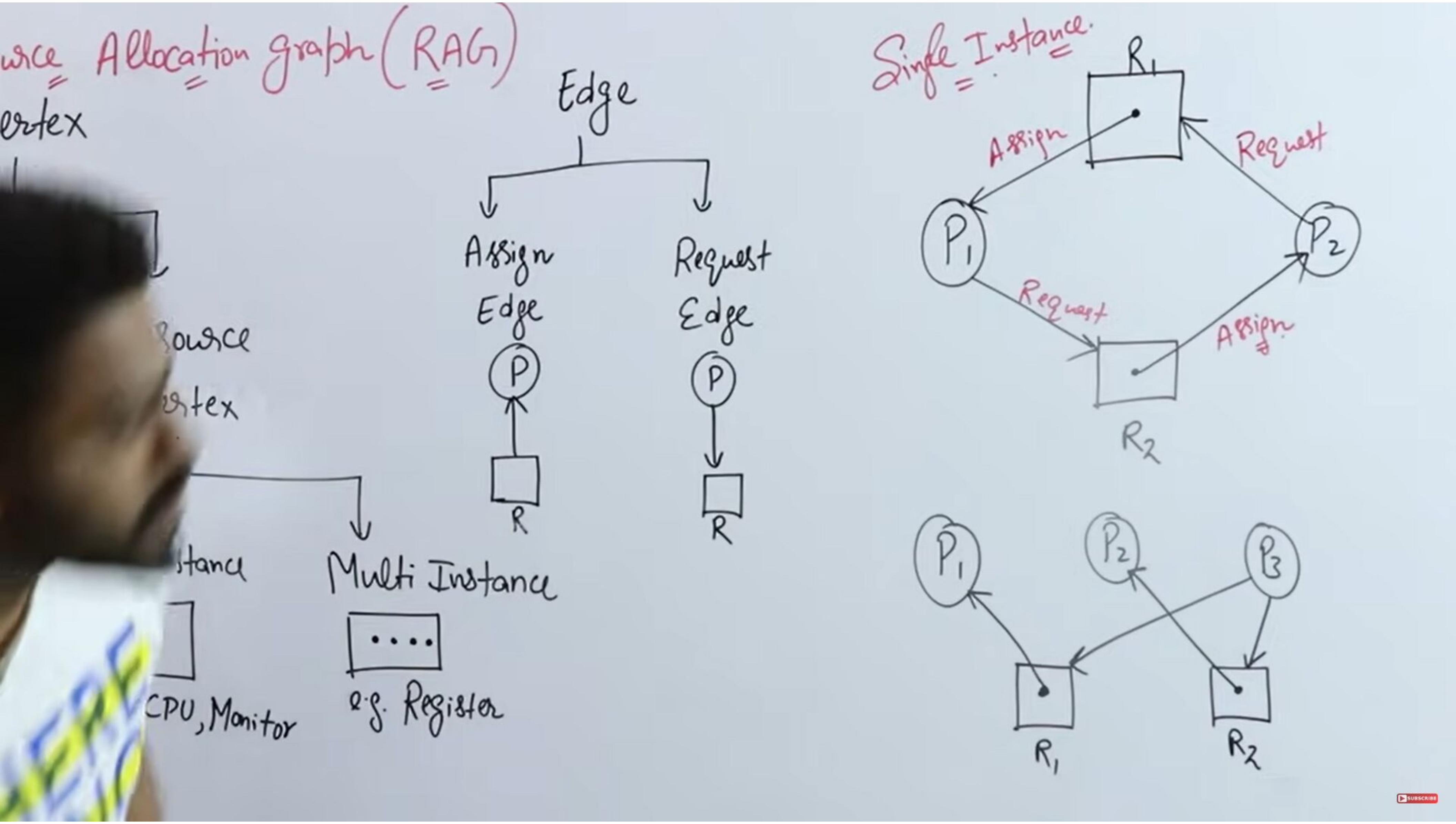


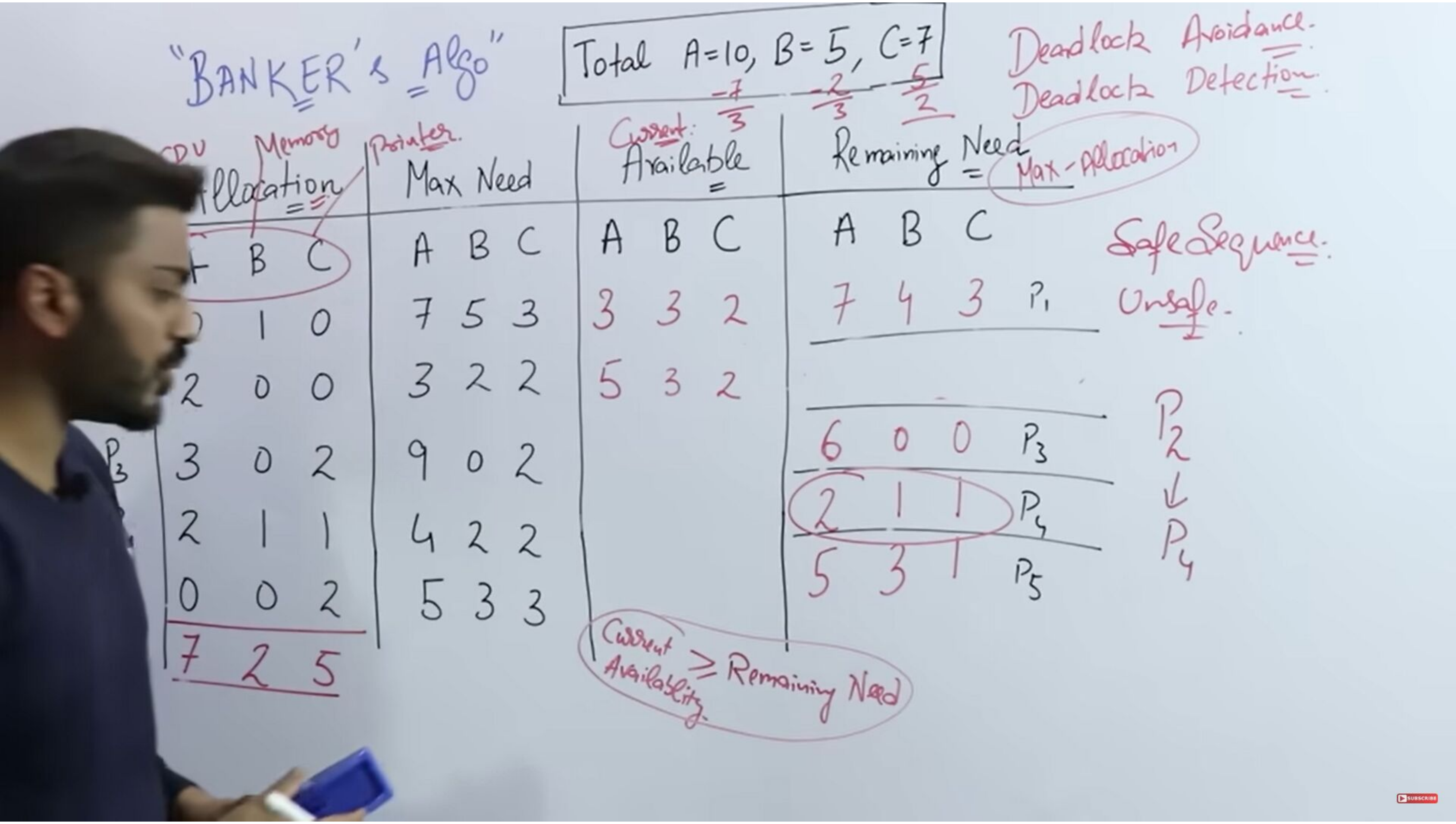


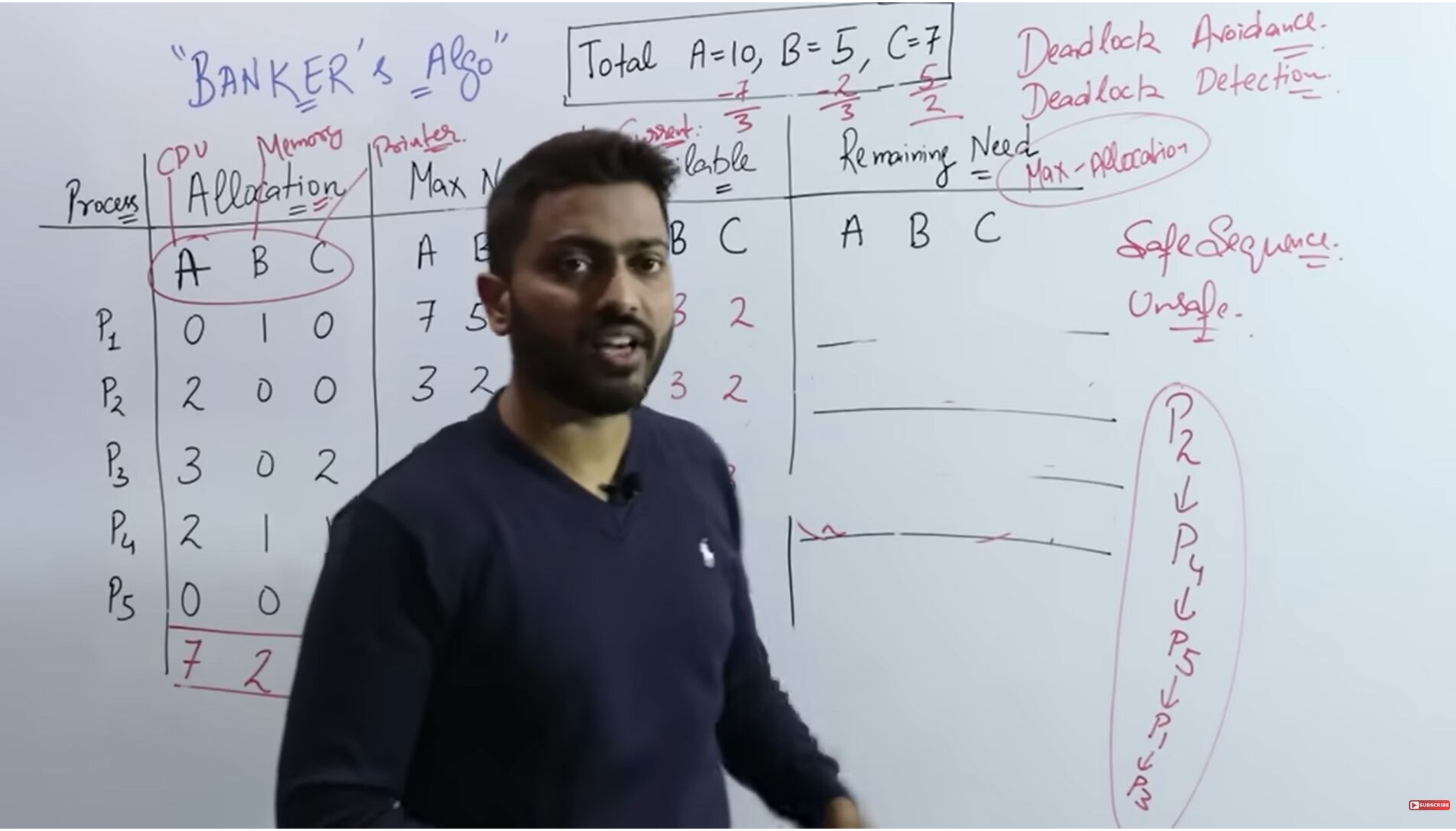

graph (RAG)
Single Instance.
Allocate
Request
R1 R2
P1 1 0 0 1 X
P2 0 1 1 0 X
Availablity (0,0)
R1
Assign
Request
P1
P2
Request
Assign
Deadlock.
Deadlock
R2
P1
P2
P3
THERE IS NO
So you can clearly say there is a deadlock in this system. So in this RAG there is a deadlock.



"Resource Allocation graph (RAG)"
Waiting.
finite
Infinite
∞
(Starvation)
1000000 years
(Deadlock)
P1 P2 P3
Allocate
Request
R1 R2
P1 1 0 0 1 X
P2 0 1 1 0 X
X P1 1 0 0 0
X P2 0 1 0 0
✓ P3 0 0 1 1
Availability (0, 0)
Single Instance.
R1
R2
Assign
Request
Deadlock.
Deadlock
P1
P2
P3


urce Allocation graph (RAG)
ertex
ource
ertex
stanc
CPU, Monitor
Multi Instance
e.g. Register
Edge
Assign Edge
Request Edge
P
R
P
R
Single Instance.
R1
Assign
Request
P1
P2
Request
Assign
R2
P1
P2
P3
R1
R2


"BANKER's Algo"
Total A=10, B=5, C=7
-7 3
-2 3
-5 2
Deadlock Avoidance.
Deadlock Detection.
CPU Memory Printer.
Allocation
A B C
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
7 2 5
Max Need
A B C
7 5 3
3 2 2
9 0 2
4 2 2
5 3 3
Current: Available
A B C
3 3 2
5 3 2
Remaining Need = Max - Allocation
A B C
7 4 3 P1
6 0 0 P3
2 1 1 P4
5 3 1 P5
Safe Sequence.
Unsafe.
P2
P4
Current Availability ≥ Remaining Need


"BANKER's Algo"
Total A=10, B=5, C=7
Deadlock Avoidance.
Deadlock Detection.
CPU
Memory
Printer.
Process
Allocation
Max N
Current
lable
Remaining Need
= Max - Allocation
A B C
P1 0 1 0
P2 2 0 0
P3 3 0 2
P4 2 1
P5 0 0
7 2
Safe Sequence.
Unsafe.
P2
P4
P5
P1
P3